CHAPTER IV

Methodology

4.1 Requirements Analysis and Requirements Documentation

Requirements analysis identified three actors: the municipal administrator, household user, and smart meter. Administrators manage accounts, devices, rates, billing, payments, requests, alerts, reports, notifications, and prioritized actions. Households monitor consumption and bills, set goals, request service, and manage preferences and access. Smart meters activate securely, retain pending telemetry, and accept server-authoritative monthly totals, limits, and leak states after matching acknowledgments.

Use Case Diagram

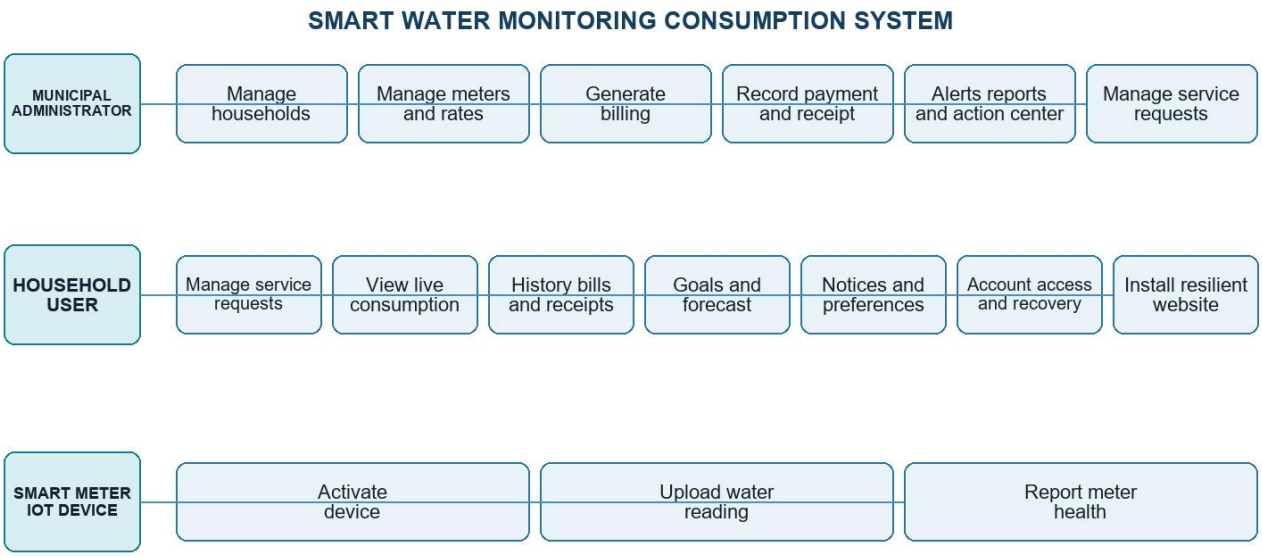


Figure 4.1. Use-case overview of the Smart Water Monitoring Consumption System.

Use Case Description

The following use-case descriptions document the implemented workflows of the municipal administrator, household user, and commissioned smart meter. Each use case includes the normal sequence, supporting paths, and exceptional conditions.

Use Case Name: Manage Households	ID:1	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to create, review, update, suspend, reactivate, or archive a household account.		
Brief Description: Maintains household identity, service address, contact information, account number, access status, and meter-assignment readiness.		
Trigger: The administrator opens Customer Management.		
Normal Flow of Events: 1. The administrator signs in and opens Customer Management. 2. The system displays searchable and filterable household records. 3. The administrator creates a new household or selects an existing record. 4. The administrator enters or updates the permitted account information. 5. The system validates required fields, unique account details, role, and status. 6. The system saves the household record, applies access changes, and logs the activity.		
Sub Flows: The administrator may unassign a meter, reset an authorized household password, or restore an eligible account.		
Alternative/Exceptional Flows: Duplicate email or account numbers, incomplete addresses, invalid statuses, and unauthorized role changes are rejected.		

Table 4.1.1 Manage Households

Use Case Name: Manage Meters and Rates	ID:2	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to manage municipal smart meters and the billing and alert rules applied to households.		
Brief Description: Combines meter registration, assignment, lifecycle control, credential management, effective-dated rates, and alert configuration.		
Trigger: The administrator opens Device Management or Water Rate Management.		
Normal Flow of Events: 1. The administrator opens the required meter or rate module. 2. The system displays registered devices or active billing configurations. 3. For a meter, the administrator registers, assigns, reassigns, suspends, or decommissions the asset. 4. For a rate, the administrator enters the water rate, fixed charge, subsidy, currency, effective date, and alert thresholds. 5. The system validates identifiers, household eligibility, numeric ranges, and effective dates. 6. The system saves the approved configuration and records the activity.		
Sub Flows: The administrator may rotate credentials, issue a new activation code, and review calibration, firmware, last-seen, and maintenance information.		
Alternative/Exceptional Flows: Duplicate assets, ineligible assignments, invalid rate values, and unauthorized changes are rejected.		

Table 4.1.2 Manage Meters and Rates

Use Case Name: Generate Billing	ID:3	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to generate a finalized monthly billing statement for an eligible household.		
Brief Description: Aggregates billable interval consumption, applies the effective configuration, and preserves a historical charge snapshot.		
Trigger: The administrator chooses Generate Statement in Billing Management.		
Normal Flow of Events: 1. The administrator selects the household and billing month. 2. The system validates the household, period, readings, and applicable rate. 3. The billing service totals liters and converts them to cubic meters. 4. The service applies the water rate, fixed charge, subsidy, and opening balance. 5. The system assigns a statement number and due date. 6. The statement, charge breakdown, balance, and payment status are stored. 7. The administrator reviews or downloads the finalized statement.		
Sub Flows: The household may download only its own finalized statement as a PDF.		
Alternative/Exceptional Flows: A duplicate month, missing billable readings, or unavailable applicable rate prevents creation of a misleading statement.		

Table 4.1.3 Generate Billing

Use Case Name: Record Offline Payment and Receipt	ID:4	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: Authorized municipal staff want to record an offline payment and issue an official receipt.		
Brief Description: Records cash, bank-deposit, check, or other offline payments without processing money through the website.		
Trigger: The administrator selects Record Payment for an unpaid or partially paid statement.		
Normal Flow of Events: 1. The administrator opens the finalized statement. 2. The administrator enters the amount, method, reference, notes, and received date. 3. The system validates the remaining balance and payment details. 4. The payment is stored in the offline-payment ledger. 5. The statement amount paid, remaining balance, and status are recalculated. 6. The system generates a unique receipt number and PDF receipt. 7. The household can download the statement and related receipts.		
Sub Flows: Multiple partial payments may be recorded until the remaining balance reaches zero.		
Alternative/Exceptional Flows: Zero, negative, or excessive amounts and payments against an ineligible statement are rejected.		

Table 4.1.4 Record Offline Payment and Receipt

Use Case Name: Review Alerts Reports and Action Center	ID:5	Important Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: The administrator wants to investigate alerts and prioritize municipal work requiring attention.		
Brief Description: Combines leak follow-up, analytics, reports, exports, failed notices, overdue statements, and the prioritized action center.		
Trigger: The administrator opens the Action Center, Leak Monitoring, Reports, Analytics, or Notifications.		
Normal Flow of Events: 1. The system combines open requests, overdue statements, leak follow-ups, and failed notices. 2. The administrator selects an item, household, report category, period, or filter. 3. The system validates authorization and input parameters. 4. Related readings, alerts, billing data, notices, and summaries are retrieved. 5. The administrator reviews evidence and records the required follow-up. 6. The administrator resolves, marks false, reopens, retries, archives, or exports the supported record.		
Sub Flows: Report previews and exports retain the selected household and period filters and write an audit entry.		
Alternative/Exceptional Flows: Invalid filters, empty export data, missing records, and unauthorized access return clear responses.		

Table 4.1.5 Review Alerts Reports and Action Center

Use Case Name: Manage Service Requests	ID:6	Important Level: High
Primary Actor: Household User; Municipal Administrator	Use Case Type: Essential, Detail	
Scenario: A household wants to request municipal assistance and staff want to track the response through resolution.		
Brief Description: Provides a referenced conversation for billing, leak, meter, account, interruption, and general service concerns.		
Trigger: The household opens Service Requests or the administrator opens the municipal request queue.		
Normal Flow of Events: 1. The household selects a category and enters the subject and description. 2. The system creates a unique request reference and an open status. 3. The household and authorized staff view the request conversation. 4. The administrator assigns priority, ownership, and status. 5. Either permitted participant adds a message; internal notes remain staff-only. 6. The administrator resolves or reopens the request as needed. 7. Updates are retained and may generate household notifications.		
Sub Flows: A request may be associated with a billing statement when the concern involves a bill.		
Alternative/Exceptional Flows: Users cannot view another household's request, and invalid transitions or internal-note access are rejected.		

Table 4.1.6 Manage Service Requests

Use Case Name: View Live Consumption	ID:7	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario: The household user wants to view the latest water consumption and meter state.		
Brief Description: Displays current flow, interval consumption, estimated billing, leak state, meter availability, and recent activity.		
Trigger: The household user opens Dashboard or Live Monitoring.		
Normal Flow of Events: 1. The household user signs in. 2. The browser requests the latest household-scoped reading and meter heartbeat. 3. The server verifies ownership and retrieves current device and consumption state. 4. The interface displays flow, liters, billing state, leak state, connection, and activity. 5. The dashboard distinguishes live, delayed, offline, missing, and invalid states. 6. The interface refreshes according to the configured polling interval.		
Sub Flows: Low-bandwidth mode reduces polling and motion while preserving explicit refresh controls.		
Alternative/Exceptional Flows: When no assigned meter or valid reading exists, the interface shows a specific unavailable state instead of a misleading zero.		

Table 4.1.7 View Live Consumption

Use Case Name: Review History and Bills	ID:8	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario: The household user wants to review past consumption and current or finalized billing information.		
Brief Description: Combines daily, weekly, monthly, or custom history with estimated billing details, statements, and receipts.		
Trigger: The user opens Consumption History or Estimated Billing.		
Normal Flow of Events: 1. The user selects the history or billing page. 2. For history, the user selects a period or valid custom date range. 3. The server retrieves only the household's readings and computes totals and chart data. 4. For billing, the server selects the applicable rate and billable interval readings. 5. If the current month is empty, the latest metered month is clearly identified. 6. The interface displays estimates, finalized statements, balances, due dates, and receipts. 7. The user downloads an owned statement or receipt when required.		
Sub Flows: Daily, weekly, monthly, and detailed history remain available without exposing another household's data.		
Alternative/Exceptional Flows: Invalid ranges, missing readings, and missing rate information remain unavailable instead of being reported as zero.		

Table 4.1.8 Review History and Bills

Use Case Name: Set Consumption Goal and Forecast	ID:9	Important Level: Medium
Primary Actor: Household User	Use Case Type: Supporting, Detail	
Scenario: The household wants to establish a monthly water-use goal and review projected end-of-month consumption.		
Brief Description: Stores one household goal per month and compares actual usage, remaining allowance, progress, and forecast.		
Trigger: The user opens Goals and Forecast and saves a target in liters.		
Normal Flow of Events: 1. The system loads the household's current-month goal and consumption. 2. The user enters or updates a positive target volume. 3. The system validates and stores the household-scoped goal. 4. Current usage is compared with the target. 5. The service estimates end-of-month consumption from valid readings. 6. The interface displays progress, remaining allowance, and forecast status.		
Sub Flows: The goal may be revised for the active month without changing other households or prior monthly records.		
Alternative/Exceptional Flows: Missing, zero, negative, or unauthorized goal values are rejected with validation feedback.		

Table 4.1.9 Set Consumption Goal and Forecast

Use Case Name: Manage Notifications and Preferences	ID:10	Important Level: Medium
Primary Actor: Household User	Use Case Type: Supporting, Detail	
Scenario: The household wants to control eligible in-app notices, language, quiet hours, and data-use preferences.		
Brief Description: Manages notification categories, read state, quiet hours, English or Filipino labels, and low-bandwidth behavior.		
Trigger: The user opens Notifications or Account Settings.		
Normal Flow of Events: 1. The system loads household-scoped notices and saved preferences. 2. The user reads, marks, or removes an eligible notice from the personal view. 3. The user selects notification categories and optional quiet hours. 4. The user selects English or Filipino and chooses low-bandwidth mode. 5. The system validates and stores the preferences. 6. The interface immediately applies supported language and data-use changes.		
Sub Flows: Critical account-security notices remain enabled, and user-side deletion does not erase the municipality's source record.		
Alternative/Exceptional Flows: Invalid quiet hours, unsupported language values, and access to another household's notices are rejected.		

Table 4.1.10 Manage Notifications and Preferences

Use Case Name: Manage Account Access and Recovery	ID:11	Important Level: High
Primary Actor: Household User	Use Case Type: Essential, Detail	
Scenario: The household wants to update contact information, change a password, or recover access.		
Brief Description: Protects household profile updates and password changes while preserving administrator-controlled account fields.		
Trigger: The user opens Account Settings or selects Forgot Password on the login page.		
Normal Flow of Events: 1. The authenticated user loads the household profile. 2. The user updates permitted contact details or submits the current and new password. 3. The system validates the request and protects role, status, and account number. 4. For recovery, the user requests a reset link for an eligible approved account. 5. The reset token and replacement password are validated. 6. The profile or password is updated and security activity is recorded.		
Sub Flows: An authorized administrator may reset a household password and revoke existing sessions.		
Alternative/Exceptional Flows: Incorrect current passwords, invalid reset tokens, ineligible accounts, and protected-field changes are rejected.		

Table 4.1.11 Manage Account Access and Recovery

Use Case Name: Install and Use Resilient Website	ID:12	Important Level: Medium
Primary Actor: Household User	Use Case Type: Supporting	
Scenario: The household wants an installable interface that remains understandable during weak or interrupted connectivity.		
Brief Description: Provides the web manifest, service worker, app-shell caching, user-scoped session snapshots, and accessible responsive controls.		
Trigger: The user installs the website or revisits it with limited connectivity.		
Normal Flow of Events: 1. The browser loads the web manifest and registers the service worker. 2. Static application-shell resources are cached for resilient loading. 3. Authenticated API responses are excluded from service-worker storage. 4. User-scoped dashboard snapshots may be retained only in session storage. 5. The interface reports offline or unavailable live state rather than showing stale data as current. 6. The user reconnects or refreshes to obtain authoritative meter information.		
Sub Flows: Theme, responsive navigation, keyboard labels, and reduced-motion behavior support accessible use across devices.		
Alternative/Exceptional Flows: Without a network connection, actions requiring current server data remain unavailable and are not silently queued as successful.		

Table 4.1.12 Install and Use Resilient Website

Use Case Name: Activate Device	ID:13	Important Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: A registered M5Stack smart meter wants to exchange a one-time code for a scoped device credential.		
Brief Description: Binds the physical hardware to the municipal device record and activates authenticated telemetry access.		
Trigger: The technician enters the administrator-issued activation code on the registered device.		
Normal Flow of Events: 1. The administrator registers the meter and issues a short-lived activation code. 2. The device submits its UUID, activation code, hardware identity, and firmware version. 3. The server verifies that the code is valid, unused, and unexpired. 4. The server checks device status and hardware-binding rules. 5. The hardware identity is bound and the device is marked active. 6. The server issues a scoped credential. 7. The M5Stack stores the credential without printing the token to the serial console.		
Sub Flows: Credential rotation or authorized reassignment requires a new activation step.		
Alternative/Exceptional Flows: Expired, reused, incorrect, or mismatched activation information is rejected and no credential is issued.		

Table 4.1.13 Activate Device

Use Case Name: Upload Water Reading	ID:14	Important Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: An activated smart meter wants to submit a measured water-consumption interval.		
Brief Description: Authenticates the device, stores the interval once, and returns the server's monthly total, water limit, and leak state.		
Trigger: Firmware version 1.1.1 reaches the configured upload interval with a pending measurement.		
Normal Flow of Events: 1. The sensor counts valid pulses and calculates flow rate and interval liters. 2. The device creates a stable reading identifier and stores the pending interval. 3. The reading is submitted using the scoped device credential. 4. The server derives the device and household from the authenticated identity. 5. The server validates and stores the interval idempotently. 6. Laravel evaluates thresholds and returns the accepted reading ID, monthly total, water limit, and leak state. 7. The device verifies that the response acknowledges the same reading ID. 8. Only then does the meter clear the pending interval and advance its sequence.		
Sub Flows: The display separates the persistent lifetime Meter value from Laravel's authoritative monthly Period value.		
Alternative/Exceptional Flows: Failed requests and malformed successful responses remain queued for retry; duplicate identifiers never double-count consumption.		

Table 4.1.14 Upload Water Reading

Use Case Name: Report Meter Health	ID:15	Important Level: Medium
Primary Actor: Smart Meter / IoT Device	Use Case Type: Supporting	
Scenario: The smart meter wants to report connection, firmware, and operational health to the server.		
Brief Description: Updates municipal visibility of connectivity, last-seen activity, telemetry state, firmware version, and diagnostics.		
Trigger: The device activates, uploads telemetry, reconnects, or reaches a health-reporting interval.		
Normal Flow of Events: 1. The device evaluates Wi-Fi state and prepares hardware and firmware information. 2. Health information accompanies authenticated telemetry where applicable. 3. The server validates the device identity and household assignment. 4. Last-seen, last-telemetry, firmware, signal, and operational fields are updated. 5. Server connectivity is derived from a successful acknowledged response. 6. The administrator workspace displays the latest device and maintenance status.		
Sub Flows: Sampling begins before network activation so setup delays do not distort the first measured interval.		
Alternative/Exceptional Flows: Invalid credentials or mismatched hardware are rejected; network loss retains pending state until reconnection.		

Table 4.1.15 Report Meter Health

4.2 System Design, Software, and Processes

The system uses an Internet of Things client-server design. Firmware version 1.1.1 on the M5Stack CoreS3 measures sensor pulses, maintains a lifetime meter value, and stores unsent interval telemetry for safe retry. The Laravel server remains authoritative for identity, ownership, the monthly billing-period total, configured water limits, leak evaluation, billing, payments, service requests, notifications, goals, and audit records. The browser provides role-specific workspaces, an installable application shell, user-scoped session snapshots, multilingual labels, theme controls, and an optional low-bandwidth mode. Authenticated API responses are not cached by the service worker, and live meter information always requires a current server response.

Output and User Interface Design

The household interface presents live readings, history, estimated and finalized billing, statements and receipts, leak information, goals and forecasts, service requests, notifications, preferences, and account controls. The administrator interface centralizes household and meter management, rates, billing and offline payments, service requests, monitoring, leak follow-up, reports, notification delivery, audit information, and the action center. Both workspaces use responsive navigation,

accessible controls, theme support, and explicit unavailable states when current data cannot be retrieved.

Figure 4.2.1 Home Interface

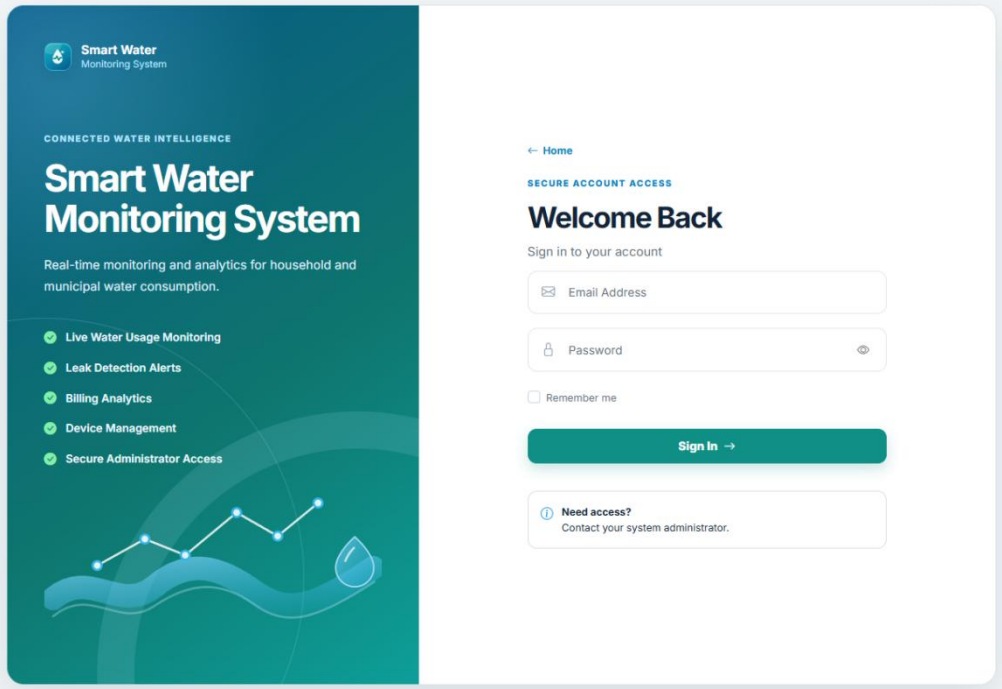


Figure 4.2.2 Log in Interface

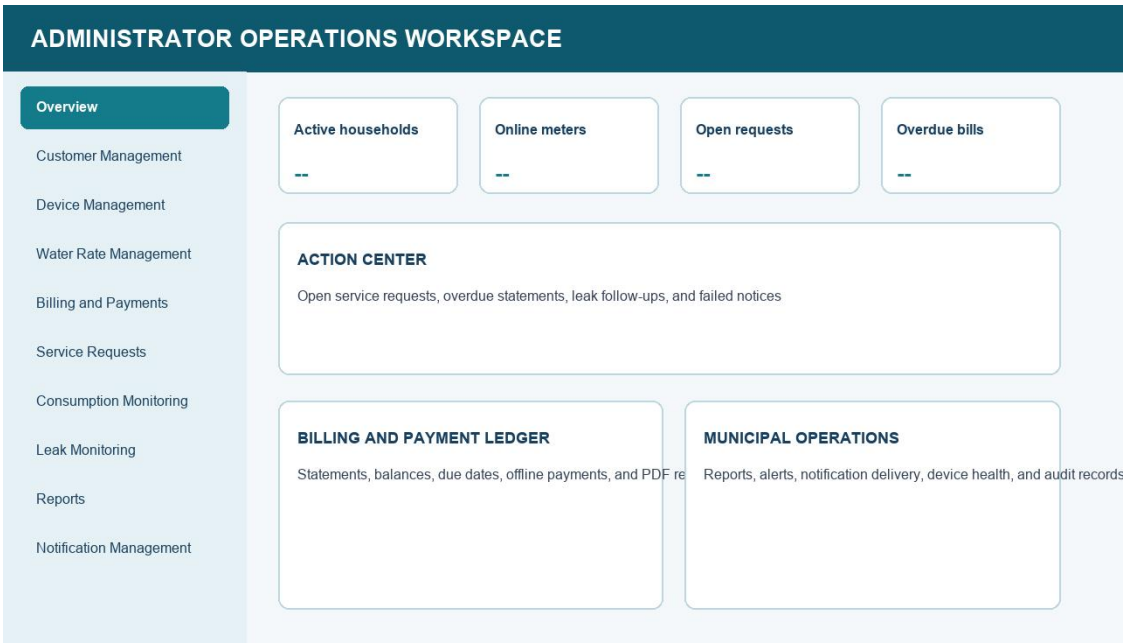


Figure 4.2.3 Updated Admin (Dashboard) Interface

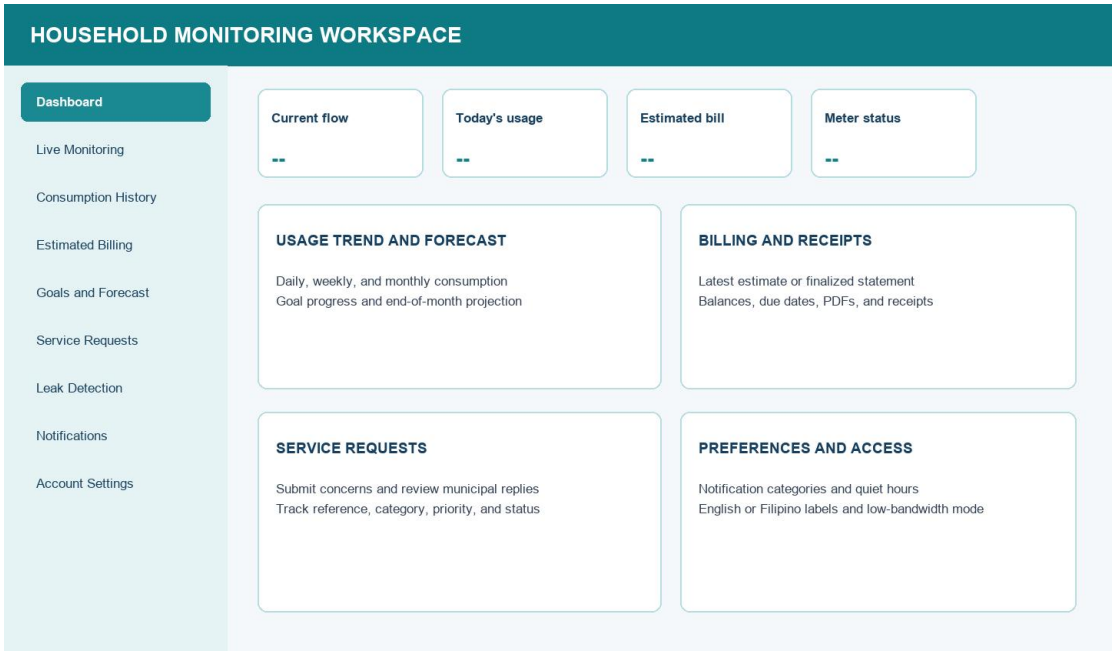
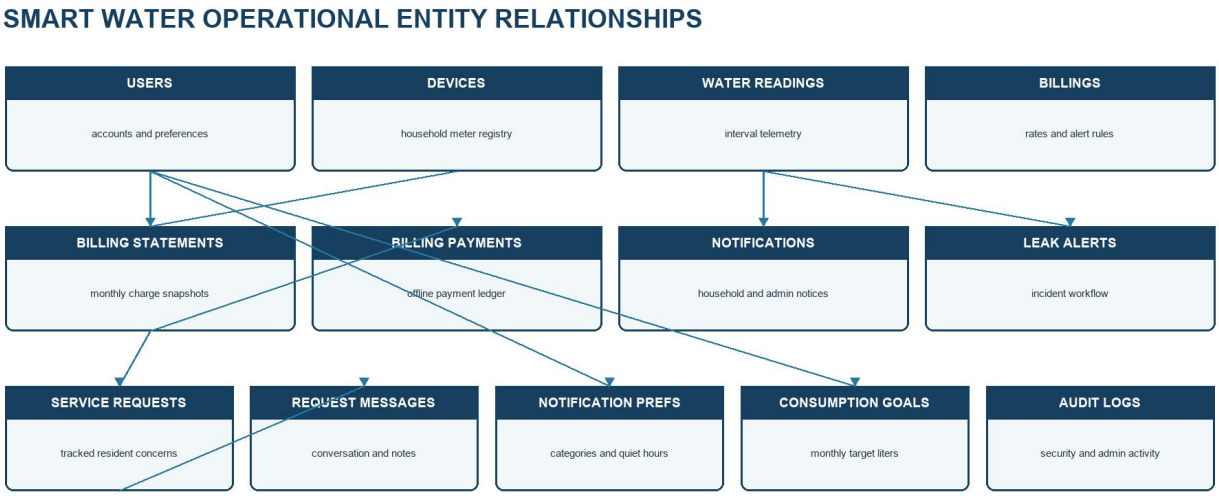


Figure 4.2.4 Updated Household (Dashboard) Interface

4.3 Entity Relationship Diagram

The operational schema links users to devices, readings, billing configurations, monthly statements, recorded payments, notices, preferences, goals, service requests, request messages, leak incidents, and audit records. The diagram summarizes the principal relationships; detailed columns are documented in Section 4.4.



4.4 Data Dictionary

The following data dictionary describes the operational tables used for monitoring, billing, resident services, preferences, alerts, and administration. Framework-support tables for cache, jobs, sessions, password resets, migrations, and personal access tokens are omitted for clarity.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the user account.

Field Name	Data Type	Length	Description
name	VARCHAR	255	Display name of the administrator, household, or device account.
email	VARCHAR	255	Unique login or account email address.
role	VARCHAR	255	Authorization role: administrator, household user, or device.
account_number	VARCHAR	255	Standardized municipal account identifier.
address	VARCHAR	255	Required service address for household accounts.
phone	VARCHAR	255	Optional contact number.
preferred_locale	VARCHAR	5	Preferred interface language: English or Filipino.
low_bandwidth_mode	BOOLEAN	1	Reduces polling and motion for lower-data use.
is_active	BOOLEAN	1	Indicates whether the account can remain operational.
email_verified_at	DATETIME	-	Date and time the email was verified.
password	VARCHAR	255	Hashed interactive-account password.
remember_token	VARCHAR	100	Framework token used for remembered sessions.
created_by_admin	BOOLEAN	1	Indicates municipal administrator provisioning.
assigned_user_id	BIGINT	20	Household linked to a legacy or supporting device account.
created_by	BIGINT	20	Administrator who created the account.
approved_at	DATETIME	-	Date and time the household account was approved.
account_status	VARCHAR	255	Lifecycle state such as active, suspended, archived, or deactivated.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.1 Database Table of Users

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the municipal device record.
uuid	VARCHAR	255	Unique server-issued device identifier.
asset_number	VARCHAR	255	Municipal equipment inventory number.
serial_number	VARCHAR	255	Unique manufacturer or field serial number.
name	VARCHAR	255	Human-readable meter name.
household_id	BIGINT	20	Currently assigned household; nullable while in inventory.
auth_user_id	BIGINT	20	Limited authentication identity associated with the device.
hardware_id	VARCHAR	255	Bound ESP32 hardware identity.
firmware_version	VARCHAR	255	Firmware version reported during activation or telemetry.
status	VARCHAR	255	Lifecycle state such as pending, active, suspended, or decommissioned.
maintenance_status	VARCHAR	255	Operational or maintenance condition.
installation_address	VARCHAR	255	Physical service location of the meter.
installed_at	DATETIME	-	Installation date and time.
last_seen_at	DATETIME	-	Latest successful device contact.
last_telemetry_at	DATETIME	-	Latest accepted telemetry timestamp.
calibrated_at	DATETIME	-	Most recent calibration date.
calibration_due_at	DATETIME	-	Scheduled calibration due date.
token_rotated_at	DATETIME	-	Latest credential rotation timestamp.
revoked_at	DATETIME	-	Credential or device revocation timestamp.
activation_code_hash	VARCHAR	255	Hash of the one-time activation code.
activation_expires_at	DATETIME	-	Expiration of the activation code.
activated_at	DATETIME	-	Successful activation timestamp.
notes	TEXT	-	Administrative installation or maintenance notes.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.2 Database Table of Devices

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the telemetry record.
user_id	BIGINT	20	Household owner of the reading.
device_id	BIGINT	20	Authenticated meter that submitted the reading.
reading_id	VARCHAR	255	Stable idempotency identifier generated by the device.
flow_rate	DECIMAL	18,6	Measured flow rate in liters per minute.
liters_used	DECIMAL	18,6	Water volume represented by the interval.
cubic_meter	DECIMAL	18,6	Consumption converted to cubic meters.
estimated_bill	DECIMAL	12,2	Estimated charge stored with the reading.
rate_per_cubic_meter	DECIMAL	12,4	Applicable water rate used for the estimate.
subsidy_deduction	DECIMAL	12,2	Subsidy applied to the estimate.
threshold_liters	DECIMAL	18,6	Consumption threshold applicable at processing time.
is_threshold_exceeded	BOOLEAN	1	Indicates consumption threshold exceedance.
is_possible_leak	BOOLEAN	1	Indicates possible leakage under server rules.
leak_detection_status	VARCHAR	255	Detailed leak evaluation state.
water_limit_liters	DECIMAL	18,6	Configured household water limit snapshot.
monitoring_status	VARCHAR	255	Device monitoring state reported with telemetry.
wifi_signal_strength	INTEGER	11	Reported Wi-Fi signal strength.
device_connection_status	VARCHAR	255	Meter-side connection state.
server_connection_status	VARCHAR	255	Latest server acknowledgement state.
device_name	VARCHAR	255	Reported or display device name.
device_payload	JSON/TEXT	-	Additional validated device diagnostics.
recorded_at	DATETIME	-	Measurement timestamp supplied or normalized by the server.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.3 Database Table of Water Readings

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the billing configuration.
user_id	BIGINT	20	Household override owner; null indicates municipal default.
water_rate	DECIMAL	12,4	Charge per cubic meter.
fixed_monthly_charge	DECIMAL	12,2	Fixed recurring monthly charge.
subsidy_deduction	DECIMAL	12,2	Configured subsidy deduction.
currency	VARCHAR	3	Three-letter billing currency, normally PHP.
effective_date	DATE	-	Date on which the configuration becomes applicable.
threshold_liters	DECIMAL	18,6	Consumption alert threshold.
consumption_alert_enabled	BOOLEAN	1	Enables consumption threshold notifications.
consumption_change_threshold_percent	DECIMAL	8,2	Percentage change from the first-month baseline that triggers an alert.
consumption_alert_unit	VARCHAR	255	Unit used by the consumption threshold.
consumption_alert_behavior	VARCHAR	255	Audience or handling rule for consumption notices.
leak_alert_enabled	BOOLEAN	1	Enables leak alert evaluation.
leak_flow_rate_threshold	DECIMAL	18,6	Flow-rate threshold used for possible leak detection.
leak_continuous_duration_minutes	INTEGER	11	Required continuous duration before leak handling.
leak_alert_behavior	VARCHAR	255	Audience or handling rule for leak notices.
quiet_hours_start	INTEGER	2	Starting hour of the configured quiet period.
quiet_hours_end	INTEGER	2	Ending hour of the configured quiet period.
is_active	BOOLEAN	1	Indicates whether the configuration is applicable.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.4 Database Table of Billing Configurations

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the monthly statement.
statement_number	VARCHAR	255	Unique municipal statement identifier.
user_id	BIGINT	20	Household billed by the statement.
billing_month	DATE	-	Month represented by the statement.
due_date	DATE	-	Payment due date shown on the finalized statement.
opening_balance	DECIMAL	12,2	Balance carried into the billing period.
total_liters	DECIMAL	18,6	Billable water consumption in liters.
total_cubic_meter	DECIMAL	18,6	Billable water consumption in cubic meters.
water_rate	DECIMAL	12,4	Rate snapshot applied at generation.
fixed_monthly_charge	DECIMAL	12,2	Fixed-charge snapshot applied at generation.
subsidy_deduction	DECIMAL	12,2	Subsidy snapshot applied at generation.
currency	VARCHAR	3	Statement currency.
current_charge	DECIMAL	12,2	Charge calculated for the current period.
amount_paid	DECIMAL	12,2	Payment recorded against the statement.
remaining_balance	DECIMAL	12,2	Outstanding balance after payment.
status	VARCHAR	255	Payment state such as unpaid, partial, or paid.
generated_at	DATETIME	-	Date and time the statement was finalized.
paid_at	DATETIME	-	Date and time full payment was recorded.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.5 Database Table of Billing Statements

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the recorded offline payment.
billing_statement_id	BIGINT	20	Finalized statement receiving the payment.
recorded_by	BIGINT	20	Administrator who recorded the payment; nullable after staff deletion.
receipt_number	VARCHAR	255	Unique official receipt identifier.
amount	DECIMAL	12,2	Amount received through an offline method.
payment_method	VARCHAR	255	Cash, bank deposit, check, or other offline method.
reference_number	VARCHAR	255	Optional external transaction or deposit reference.
notes	TEXT	-	Optional municipal payment notes.
received_at	DATETIME	-	Date and time the payment was received.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.6 Database Table of Billing Payments

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the notification.
user_id	BIGINT	20	Household recipient; nullable for administrative or batch scope.
water_reading_id	BIGINT	20	Related water reading when the notice is telemetry-driven.
type	VARCHAR	255	Notification category.
title	VARCHAR	255	Short notification heading.
message	TEXT	-	Full notification content.
metadata	JSON/TEXT	-	Structured context used by the interface or workflow.
is_read	BOOLEAN	1	Indicates whether the recipient opened the notice.
batch_id	VARCHAR	255	Identifier shared by a batch announcement.
delivery_status	VARCHAR	255	Delivery lifecycle state.
delivery_error	TEXT	-	Latest delivery failure message when applicable.
scheduled_at	DATETIME	-	Scheduled delivery timestamp.
sent_at	DATETIME	-	Actual delivery or creation timestamp.
archived_at	DATETIME	-	Administrative archive timestamp.
user_deleted_at	DATETIME	-	Household-side removal timestamp without source deletion.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.7 Database Table of Notifications

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the household preference record.
user_id	BIGINT	20	Unique household that owns the preferences.
announcements	BOOLEAN	1	Enables general municipal announcements.
billing_updates	BOOLEAN	1	Enables billing-related updates.
consumption_alerts	BOOLEAN	1	Enables consumption-change notifications.
leak_alerts	BOOLEAN	1	Enables leak-related notifications.
service_request_updates	BOOLEAN	1	Enables service-request status and reply notifications.
quiet_hours_start	TIME	-	Optional start of the quiet-hours window.
quiet_hours_end	TIME	-	Optional end of the quiet-hours window.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.8 Database Table of Notification Preferences

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the monthly goal.
user_id	BIGINT	20	Household that owns the goal.
month	DATE	-	Month to which the goal applies.
target_liters	DECIMAL	12,2	Household target consumption in liters.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.9 Database Table of Consumption Goals

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the service request.
reference_number	VARCHAR	255	Unique household-facing request reference.
user_id	BIGINT	20	Household that owns the request.
assigned_to	BIGINT	20	Administrator assigned to the request; nullable.
billing_statement_id	BIGINT	20	Optional related billing statement.
category	VARCHAR	255	Billing, leak, account, interruption, meter, or general category.
subject	VARCHAR	255	Short description supplied by the household.
description	TEXT	-	Initial detailed concern or request.
priority	VARCHAR	255	Municipal priority such as normal, high, or urgent.
status	VARCHAR	255	Workflow state such as open, in progress, resolved, or closed.
resolved_at	DATETIME	-	Date and time the request was resolved.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.10 Database Table of Service Requests

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the request message.
service_request_id	BIGINT	20	Service request containing the message.
user_id	BIGINT	20	Household or administrator who authored the message.
message	TEXT	-	Conversation message or staff note.
is_internal	BOOLEAN	1	Indicates a staff-only message hidden from the household.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.11 Database Table of Service Request Messages

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the leak incident.
user_id	BIGINT	20	Affected household.
first_water_reading_id	BIGINT	20	First reading associated with the incident.
latest_water_reading_id	BIGINT	20	Most recent supporting reading.
notification_id	BIGINT	20	Notification associated with the incident.
status	VARCHAR	255	Incident state such as detected, acknowledged, resolved, or false alarm.
notes	TEXT	-	Administrative investigation and resolution notes.
detected_at	DATETIME	-	Initial detection timestamp.
last_detected_at	DATETIME	-	Most recent detection timestamp.
notified_at	DATETIME	-	Household notification timestamp.
acknowledged_at	DATETIME	-	Administrative acknowledgement timestamp.
resolved_at	DATETIME	-	Resolution timestamp.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.12 Database Table of Leak Alerts

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the audit record.
user_id	BIGINT	20	Authenticated actor when available.
action	VARCHAR	255	Machine-readable administrative or security action.
auditable_type	VARCHAR	255	Type of affected domain record.
auditable_id	BIGINT	20	Identifier of the affected domain record.
description	VARCHAR	255	Human-readable activity description.
metadata	JSON/TEXT	-	Redacted structured activity context.
outcome	VARCHAR	255	Result such as success or failure.
ip_address	VARCHAR	45	Request IP address when retained.
user_agent	TEXT	-	Request client or browser description.
request_id	VARCHAR	255	Request correlation identifier.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.13 Database Table of Audit Logs

4.5 Development and Testing

Development and verification covered authentication and account restrictions; household and meter management; device activation, credential rotation, reassignment, and health; idempotent telemetry; server-authoritative monthly totals and water limits; live and historical consumption; billing estimates, statements, offline payments, and PDF receipts; leak workflows; service requests; goals and forecasts; notification preferences; reports and exports; PWA and accessibility behavior; and security controls. On September 9, 2026, the complete Laravel test suite passed 166 tests with 1,096 assertions. The Vite production build also completed successfully, producing optimized CSS and JavaScript assets. Physical deployment should additionally record calibrated volume tests and network field observations for the installed sensor and meter assembly.

4.6 Implementation Plan

Implementation begins by preparing the Laravel environment, database, application URL, mail configuration, production assets, and municipal administrator account. Staff then configure effective water rates and alert rules, register household accounts, register and assign each meter, and issue a short-lived activation code. Firmware version 1.1.1 is configured with Wi-Fi and server information, uploaded to the M5Stack CoreS3, and verified through hardware binding, last-seen status, acknowledged readings, monthly Period totals, and the lifetime Meter display. After telemetry is confirmed, staff validate dashboards, statements and receipts, service-request routing, notifications and preferences, goals and forecasts, exports, action-center items, PWA installation, and recovery procedures before operational release.

Implementation Issues and Challenges

Implementation challenges included incorrect wiring, defective sensor components, unstable pulse signals, network errors, duplicated or malformed telemetry responses, month-boundary billing behavior, and the risk of presenting missing data as zero. Hardware issues were addressed through component and wiring checks, calibration, pulse-noise filtering, and repeated volume tests. Software issues were addressed through stable reading identifiers, acknowledgement matching, retained pending intervals, server-derived limits and connection state, current-period aggregation, latest-metered-month fallback, explicit unavailable states, validation, audit logging, and automated regression tests.

4.7 System Implementation and Results

The updated Smart Water Monitoring System implements the complete path from measured water flow to household information and municipal action. The M5Stack CoreS3 records sensor pulses, maintains a lifetime Meter value, and submits retry-safe interval readings. Laravel authenticates the commissioned device, stores each reading once, calculates the authoritative monthly Period total, applies water limits and leak rules, and supports current and historical consumption, estimated and finalized billing, offline payments, statements and receipts, service requests, notifications and preferences, goals and forecasts, reports, exports, and audit records. Household and administrator workspaces expose these functions through responsive, accessible, theme-aware interfaces with PWA and low-bandwidth behavior. The successful 166-test, 1,096-assertion verification run and production asset build confirm the implemented software workflows; calibrated

hardware accuracy should remain documented through deployment-specific field measurements.